

Foundation Models for X-ray Scattering Image Attribute Classification

1 Motivation

The X-Ray from the Synchrotron are used in studying the physical structure of materials by scattering them around probe objects and then studying the pattern in the resulting x-ray images. Modern detectors produce images in the order of millions of images per day, necessitating an automatic analysis of these images.

This project proposes a pipeline that would enable automatic classification of these generated images into pre-defined attribute types using *foundation models* (FM). Although there are millions of images available, the annotated images are very limited because of the need for significant manual effort in annotating the attributes. Hence the project plans to use the *Self Supervised Learning* (SSL) capabilities of foundation models to capture key image features by leveraging the large set of unlabelled images.

Dinov2 has shown to adapt well in the medical domain as seen in [3][4][5] and thus is used as a base. The idea of building a FM with novel architecture of X ray scattering is left in the future work due to data, compute, time and expertise requirements.

2 Proposed System Overview

Stage	Input	Method
SSL pre-training	50–100 k unlabeled patterns (plus synthetic)	DINOv2 ViT-L with momentum teacher and multi-crop strategy
Supervised fine-tuning	~3 k labeled images	LoRA adapters, class-balanced focal loss
Fast baseline	Labeled set + Synthetic	ResNet-34 (scratch), confidence threshold $T = 0.5$
Inference	New experimental runs (1000s images)	(1) CNN baseline; (2) FM on low-confidence cases

The pipeline would consist of using a pretrained DINOv2 ViT model and pretraining on decently large unlabelled x-ray scattering images from the synchrotron. Then we will use performance efficient fine tuning with adapters on our labeled dataset. We will also train a ResNet from scratch using our labeled and synthetic dataset that will function as a baseline. In the full workflow, we will run all the images on the ResNet to classify the images and route the low confidence cases to the foundation model.

3 Detailed Plan

3.1 Data Pipeline

- **Collection:** Annotated data from [2], Synthetic data from [1], Unlabelled data from Lab experiments, public SAXS/WAXS archives, synthetic Debye simulations.
- **Pre-processing:** *Normal:* bit-depth standardization, resize, crop, rotations/flips, per-image z-score, Gaussian noise augmentation, *X-ray-scattering-specific:* beam-stop masking, dead-pixel correction, radial q-scaling, logarithmic intensity, background subtraction, per-ring normalization, Poisson noise augmentation.

3.2 Modelling Approaches

Primary Foundation Model route

- Take the DINOv2 model weights and continue the self Supervised learning with the self distillation method as outlined in [7] for 5 epochs on the large unlabelled dataset. FSDP or DeepSpeed with ZeRo optimizer for the implementation part with mixed precision would enable faster training by splitting into multiple parallel processes - Distributed Training.
- Fine-tune via LoRA on labelled dataset; freeze 90 % of backbone; optimise BCE with focal loss; early-stop.

Baselines / Ablations

- ResNet-34 baseline for speed.
- Out-of-box DINOv2 + linear head (value of extra SSL).
- SimCLR pre-training with ResNet-50.

Extensions

- MAE pre-training for texture-rich diffraction rings in place of DINOv2
- CLIP-style multimodal model pairing images with textual metadata to leverage image text coreferences present in literature
- Pretrained encoding part of diffusion based models like Stable Diffusion
- SAM-style segmentation FM for peak localisation.

3.3 Training Logistics

- Cluster: 8 - A100 80 GB; throughput - 720 img/s; Sharded WebDataset on S3; wandb tracking.

3.4 Evaluation Metrics

Phase	Metric	Rationale
SSL	k-means purity, 1-NN top-1	Representation quality
Fine-tune	macro-mAP, per-label F1	Handles class imbalance
Production	latency/img, GPU-hr/1 k imgs	Deployment cost

3.5 Deployment Strategy

We will implement a two-stage cascade model for better efficiency:

- ResNet baseline (- 3 ms/img GPU); FM only for less confident scores- softmax $< \tau$ (- 15 ms/img).
- INT8 quantisation and adapter pruning (30 %) cut memory if speed and efficiency are the bottlenecks
- Active learning loop to collect highscoring samples at run time to add to the training data as soft inputs - Semi Supervised learning.

4 Future Work

- Use GRPO to optimize a physics-aware reward function to penalize labels violating scattering principles computing group-relative advantages to reinforce physically plausible classifications.
- DiT based rare pattern synthesis to generate high resolution samples of rare patterns to be used in training.
- Build a novel FM architecture from scratch by using physics informed neural network modules into each transformer that enforces physical equations as soft constraints and also accounts for scattering specific activations, rather than learn from statistical distribution of image features here.

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